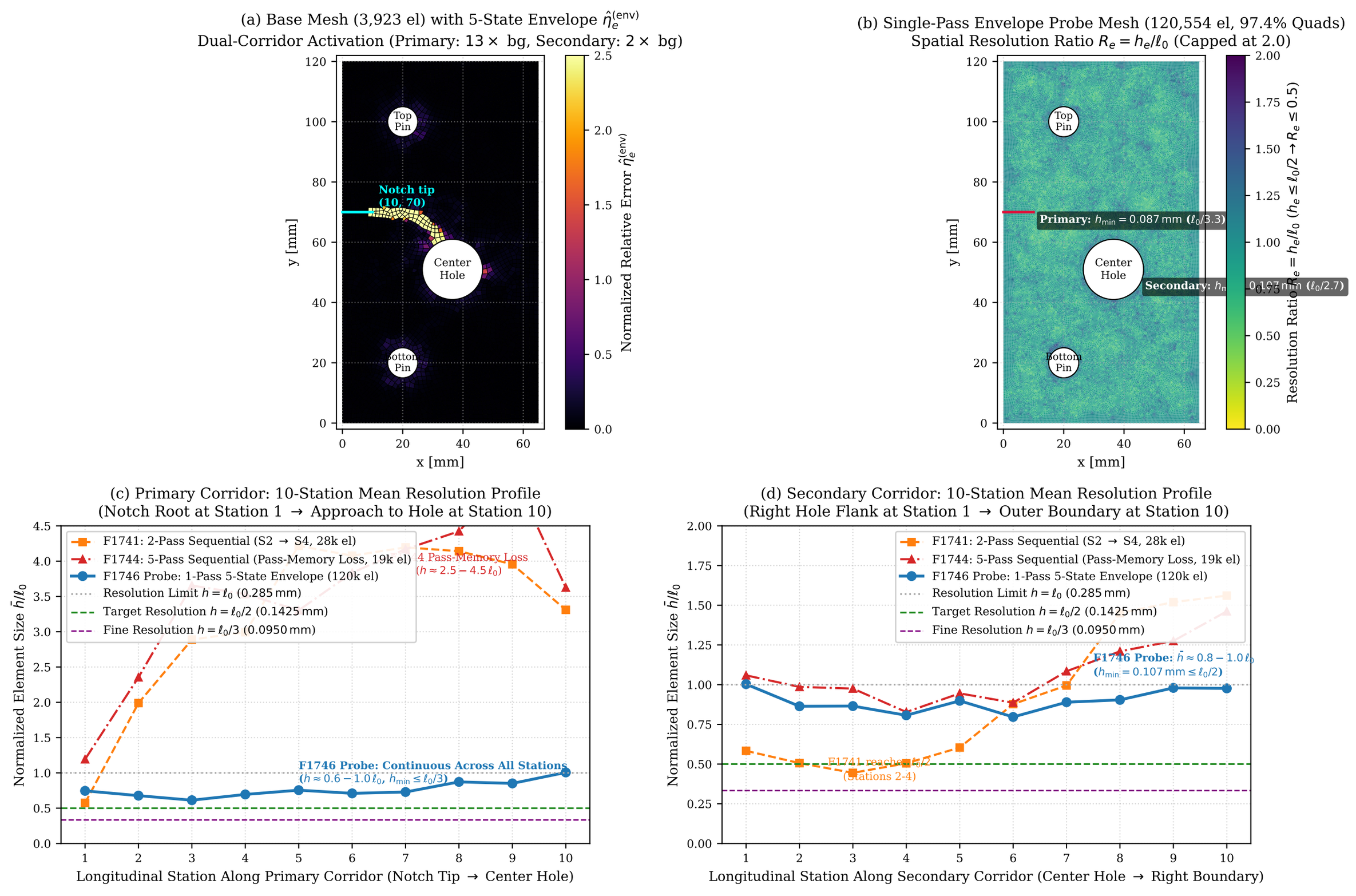

(a) Base Mesh (3,923 el) with 5-State Envelope $\hat{\eta}_e^{(\mathrm{env})}$
Dual-Corridor Activation (Primary: 13 × bg, Secondary: 2 × bg)
Top Pin
Notch tip (10, 70)
Center Hole
Bottom Pin
x [mm]
y [mm]
Normalized Relative Error $\hat{\eta}_e^{(\mathrm{env})}$
(b) Single-Pass Envelope Probe Mesh (120,554 el, 97.4% Quads)
Spatial Resolution Ratio $R_e = h_e/\ell_0$ (Capped at 2.0)
Primary: $h_{\min}$ = 0.087 mm (ℓ_0/3.3)
Secondary: $h_{\min}$ = 0.107 mm (ℓ_0/2.7)
Resolution Ratio $R_e = h_e/\ell_0$ ($h_e \le \ell_0/2 \to R_e \le 0.5$)
(c) Primary Corridor: 10-Station Mean Resolution Profile
(Notch Root at Station 1 → Approach to Hole at Station 10)
F1741: 2-Pass Sequential (S2 → S4, 28k el)
F1744: 5-Pass Sequential (Pass-Memory Loss, 19k el)
F1746 Probe: 1-Pass 5-State Envelope (120k el)
Resolution Limit $h = \ell_0$ (0.285 mm)
Target Resolution $h = \ell_0/2$ (0.1425 mm)
Fine Resolution $h = \ell_0/3$ (0.0950 mm)
4 Pass-Memory Loss ($h \approx 2.5 - 4.5\,\ell_0$)
F1746 Probe: Continuous Across All Stations ($h \approx 0.6 - 1.0\,\ell_0$, $h_{\min} \le \ell_0/3$)
Normalized Element Size $\bar{h}/\ell_0$
Longitudinal Station Along Primary Corridor (Notch Tip → Center Hole)
(d) Secondary Corridor: 10-Station Mean Resolution Profile
(Right Hole Flank at Station 1 → Outer Boundary at Station 10)
F1746 Probe: $\bar{h} \approx 0.8 - 1.0\,\ell_0$ ($h_{\min} = 0.107$ mm $\le \ell_0/2$)
F1741 reaches $\ell_0/2$ (Stations 2-4)
Longitudinal Station Along Secondary Corridor (Center Hole → Right Boundary)